Mobile Application Development (Android)

Complete Laboratory Practicals Manual — Source Code, Layouts & Activity Implementations

This document contains the complete, verified source code, XML layouts, and architectural explanations for all 10 Mobile Application Development practicals, extracted and formatted for easy study, deployment, and version control on GitHub.

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GitHub Setup & Repository Deployment

To initialize a Git repository and push all practical files to your personal GitHub account:

```
# 1. Navigate to the extracted repository folder
cd android_practicals

# 2. Initialize a Git repository
git init

# 3. Add all files to staging
git add .

# 4. Commit the practicals
git commit -m "Add complete Android Mobile Application Development practicals"

# 5. Set default branch to main
git branch -M main

# 6. Link to your GitHub remote repository (replace with your repo URL)
git remote add origin https://github.com/<your-username>/<your-repo-name>.git

# 7. Push to GitHub
git push -u origin main
```

Detailed Practicals & Complete Source Code

Practical 01: Introduction to Android and Android Studio IDE

Aim: Introduction to Android, Introduction to Android Studio IDE.

Overview: Conceptual practical covering the Android open-source Linux-based architecture and IDE tooling. No code is required for this practical.

- **Activity:** A single user-facing screen or user interface responsible for user interactions (e.g., Login, Home, Profile).
- **Service:** Background worker performing long-running operations without providing a user interface (e.g., playing audio, network sync).
- **Broadcast Receiver:** Listens and reacts to system-wide or custom application broadcast events (e.g., low battery, boot completed).
- **Content Provider:** Manages shared application datasets securely across applications (e.g., Contacts, MediaStore).
- **Android Studio IDE:** Official IDE providing Code Editor, Layout Editor, Android Emulator (AVD), Logcat, Debugger, and Gradle build engine.

Practical 02: Simple "Hello World" Program

Aim: Creating a Project, Android Components, Activities, Services, Content Providers, Broadcast Receivers, Interface overview, Creating Android Virtual device, USB debugging mode, Android Application Overview. Simple “Hello World” program.

Activity Role: `MainActivity.java` (Package: `com.example.helloworld`) serves as the primary entry point activity. Extending `AppCompatActivity`, it overrides `onCreate()` to initialize the activity state and render the visual layout via `setContentView(R.layout.activity_main)`.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World"
        android:textSize="28sp" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Practical 03: Android Resources

Aim: Programming Resources: Android Resources: (Color, Theme, String, Drawable, Dimension, Image).

Activity Role: `MainActivity.java` (Package: `com.example.aimlprac2`) loads `activity_main.xml`, which demonstrates declarative separation of UI assets by referencing colors, strings, themes, dimensions, and drawables stored under `res/`.

1. res/values/colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="blue">#2196F3</color>
    <color name="red">#F44336</color>
    <color name="white">#FFFFFF</color>
</resources>
```

2. res/values/strings.xml

```
<resources>
    <string name="app_name">Resource Demo</string>
    <string name="welcome">Welcome to Android</string>
    <string name="button_text">Login</string>
</resources>
```

3. res/values/themes.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="Theme.Aimlprac2"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
        <item name="colorPrimary">@color/blue</item>
        <item name="colorPrimaryVariant">@color/red</item>
        <item name="colorOnPrimary">@color/white</item>
    </style>
</resources>
```

4. res/values/dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <dimen name="padding">20dp</dimen>
    <dimen name="text_size">24sp</dimen>
    <dimen name="image_size">150dp</dimen>
</resources>
```

5. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Android"
        android:textColor="@color/red"
        android:textSize="24sp" />

    <ImageView
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:src="@drawable/logo" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login" />
```

```
</LinearLayout>
```

6. MainActivity.java

```
package com.example.aimlprac2;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Practical 04: Activity Life Cycle and Activity Methods

Aim: Programming Activities and fragments : Activity Life Cycle, Activity methods.

Activity Role: `MainActivity.java` (Package: `com.example.activitylifecycledemo`) intercepts and logs every stage of the Activity lifecycle (`onCreate`, `onStart`, `onResume`, `onPause`, `onStop`, `onRestart`, and `onDestroy`). It triggers both visual `Toast` alerts and debug logs in `Logcat` under tag `"ActivityLifeCycle"`.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Life Cycle Demo"
        android:textSize="24sp" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.activitylifecycledemo;

import android.os.Bundle;
import android.util.Log;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Toast.makeText(this, "onCreate()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onCreate()");
    }

    @Override
    protected void onStart() {
        super.onStart();
        Toast.makeText(this, "onStart()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onStart()");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(this, "onResume()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onResume()");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(this, "onPause()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onPause()");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Toast.makeText(this, "onStop()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onStop()");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Toast.makeText(this, "onRestart()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onRestart()");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Toast.makeText(this, "onDestroy()", Toast.LENGTH_SHORT).show();
        Log.d("ActivityLifeCycle", "onDestroy()");
    }
}
```

Practical 05: Android Layouts, ListView, and GridView

Aim: Programs related to different Layouts : Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.

Activity Role: `MainActivity.java` (Package: `com.example.aiml5prc`) wires data adapters to two dynamic adapter components: a `ListView` (rendering fruits: Apple, Banana, Orange, Mango, Grapes) and a 3-column `GridView` (rendering letters: A, B, C, D, E, F) using `ArrayAdapter<String>`.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <!-- 1. Linear Layout -->
        <TextView
            android:text="1. Linear Layout"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <LinearLayout
            android:orientation="horizontal"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="8dp"
            android:layout_marginBottom="16dp">
            <Button android:text="Button 1" android:layout_width="wrap_content"
android:layout_height="wrap_content" />
            <Button android:text="Button 2" android:layout_marginStart="8dp"
android:layout_width="wrap_content" android:layout_height="wrap_content" />
        </LinearLayout>

        <!-- 2. Relative Layout -->
        <TextView
            android:text="2. Relative Layout"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <RelativeLayout
            android:layout_width="match_parent"
            android:layout_height="70dp"
            android:layout_marginBottom="16dp">
            <Button android:text="Center" android:layout_centerInParent="true"
android:layout_width="wrap_content" android:layout_height="wrap_content" />
        </RelativeLayout>

        <!-- 3. Table Layout -->
        <TextView
            android:text="3. Table Layout"
            android:textStyle="bold"
```

```
            android:textSize="18sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <TableLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginBottom="16dp">
            <TableRow>
                <TextView android:text="Name " android:textSize="16sp" />
                <TextView android:text="RAMAN" android:textSize="16sp"
android:textStyle="bold" />
            </TableRow>
            <TableRow>
                <TextView android:text="Age " android:textSize="16sp" />
                <TextView android:text="24" android:textSize="16sp" />
            </TableRow>
        </TableLayout>

        <!-- 4. Frame Layout -->
        <TextView
            android:text="4. Frame Layout"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <FrameLayout
            android:background="#E0E0E0"
            android:layout_width="160dp"
            android:layout_height="70dp"
            android:layout_marginBottom="16dp">
            <TextView android:text="Frame Layout" android:layout_gravity="center"
android:layout_width="wrap_content" android:layout_height="wrap_content" />
        </FrameLayout>

        <!-- 5. Absolute / Coordinate Layout -->
        <TextView
            android:text="5. Absolute / Coordinate Layout"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <RelativeLayout
            android:layout_width="match_parent"
            android:layout_height="70dp"
            android:layout_marginBottom="16dp">
            <Button android:text="Positioned" android:layout_marginStart="40dp"
android:layout_width="wrap_content" android:layout_height="wrap_content" />
        </RelativeLayout>

        <!-- 6. ListView -->
        <TextView
            android:text="6. ListView"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />
```

```
        <ListView
            android:id="@+id/listView"
            android:layout_width="match_parent"
            android:layout_height="250dp" />

        <!-- 7. GridView -->
        <TextView
            android:text="7. GridView"
            android:textStyle="bold"
            android:textSize="18sp"
            android:layout_marginTop="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content" />

        <GridView
            android:id="@+id/gridView"
            android:layout_width="match_parent"
            android:layout_height="150dp"
            android:numColumns="3" />

    </LinearLayout>
</ScrollView>
```

2. MainActivity.java

```
package com.example.aiml5prc;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listView;
    GridView gridView;

    String[] listItems = {
        "Apple", "Banana", "Orange", "Mango", "Grapes"
    };

    String[] gridItems = {
        "A", "B", "C", "D", "E", "F"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);
        gridView = findViewById(R.id.gridView);

        ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
```

```
            listItems
        );
        listView.setAdapter(listAdapter);

        ArrayAdapter<String> gridAdapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            gridItems
        );
        gridView.setAdapter(gridAdapter);
    }
}
```

Practical 06: UI Elements, AppBar, and Fragments

Aim: Programming UI elements:AppBar, Fragments, UI Components.

Activity & Fragment Roles:

- `MainActivity.java` (Package: `com.example.aimlprac6`): Configures the modern `Toolbar` widget as the activity's action bar using `setSupportActionBar()` and performs a dynamic `FragmentTransaction` to place `MyFragment` inside the `fragment_container` FrameLayout.
- `MyFragment.java` (Package: `com.example.aimlprac6`): Subclasses `Fragment`, inflates `fragment_my.xml` (with input controls: EditText, CheckBoxes, RadioGroup, and Switch), and displays a greeting Toast upon clicking the Submit button.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="#6200EE"
        app:title="UI Elements Demo"
        app:titleTextColor="@android:color/white" />

    <FrameLayout
        android:id="@+id/fragment_container"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.aimlprac6;
```

```
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

    Toolbar toolbar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);

        getSupportFragmentManager()
            .beginTransaction()
            .replace(R.id.fragment_container, new MyFragment())
            .commit();
    }
}
```

3. res/layout/fragment_my.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Student Information"
            android:textSize="24sp"
            android:textStyle="bold" />

        <EditText
            android:id="@+id/etName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Name" />

        <Button
            android:id="@+id/btnSubmit"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Submit" />

        <CheckBox
            android:id="@+id/check1"
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Java" />

        <CheckBox
            android:id="@+id/check2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Android" />

        <RadioGroup
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">
            <RadioButton android:id="@+id/r1" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Male" />
            <RadioButton android:id="@+id/r2" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Female" />
        </RadioGroup>

        <Switch
            android:id="@+id/switch1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Enable Notification" />

    </LinearLayout>
</ScrollView>
```

4. MyFragment.java

```
package com.example.aimlprac6;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class MyFragment extends Fragment {

    EditText etName;
    Button btnSubmit;

    public MyFragment() {
    }

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        View view = inflater.inflate(R.layout.fragment_my, container, false);
```

```
        etName = view.findViewById(R.id.etName);
        btnSubmit = view.findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(v -> {
            String name = etName.getText().toString();
            Toast.makeText(getActivity(), "Welcome " + name, Toast.LENGTH_SHORT).show();
        });

        return view;
    }
}
```

Practical 07: Menus, Dialogs, and DialogFragments

Aim: Programming menus, dialog, dialog fragments.

Activity & DialogFragment Roles:

- `MainActivity.java` (Package: `com.example.aimlprac7`): Displays an `AlertDialog` with confirmation buttons on `btnDialog` click, instantiates and shows `MyDialogFragment` on `btnFragment` click, inflates the options menu in `onCreateOptionsMenu()`, and dispatches menu action clicks in `onOptionsItemSelected()`.
- `MyDialogFragment.java` (Package: `com.example.aimlprac7`): Encapsulates an `AlertDialog` inside a lifecycle-managed `DialogFragment`.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu, Dialog and DialogFragment"
        android:textSize="20sp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/btnDialog"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:text="Show Dialog" />

    <Button
        android:id="@+id/btnFragment"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:layout_marginTop="15dp"
        android:text="Show Dialog Fragment" />

</LinearLayout>
```

2. res/menu/main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:id="@+id/menu_home" android:title="Home" />
    <item android:id="@+id/menu_settings" android:title="Settings" />
    <item android:id="@+id/menu_exit" android:title="Exit" />
</menu>
```

3. MainActivity.java

```
package com.example.aimlprac7;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnDialog, btnFragment;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnDialog = findViewById(R.id.btnDialog);
        btnFragment = findViewById(R.id.btnFragment);

        // Alert Dialog
        btnDialog.setOnClickListener(v -> {
            new AlertDialog.Builder(MainActivity.this)
                .setTitle("Confirmation")
                .setMessage("Do you want to continue?")
                .setPositiveButton("Yes", (dialog, which) -> {
                    Toast.makeText(MainActivity.this, "You clicked Yes",
Toast.LENGTH_SHORT).show();
                })
                .setNegativeButton("No", null)
                .show();
        });

        // Dialog Fragment
        btnFragment.setOnClickListener(v -> {
            MyDialogFragment dialogFragment = new MyDialogFragment();
            dialogFragment.show(getSupportFragmentManager(), "MyDialog");
        });
    }
```

```
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        if (item.getItemId() == R.id.menu_home) {
            Toast.makeText(this, "Home Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (item.getItemId() == R.id.menu_settings) {
            Toast.makeText(this, "Settings Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (item.getItemId() == R.id.menu_exit) {
            finish();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
}
```

4. MyDialogFragment.java

```
package com.example.aimlprac7;

import android.app.Dialog;
import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.DialogFragment;

public class MyDialogFragment extends DialogFragment {

    @NonNull
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        return new AlertDialog.Builder(requireContext())
            .setTitle("Dialog Fragment")
            .setMessage("This is a DialogFragment.")
            .setPositiveButton("OK", null)
            .setNegativeButton("Cancel", null)
            .create();
    }
}
```

Practical 08: Intents, Events, Listeners, and Adapters

Aim: Programs on Intents, Events, Listeners and Adapters: The Android Intent Class, Using Events and Event Listeners.

Activity Roles:

- `MainActivity.java` (Package: `com.example.intenteventadapterdemo`): Implements an `OnClickListener` on a button that constructs an explicit `Intent` targeting `SecondActivity.class` and launches it using `startActivity()`.

- `SecondActivity.java` (Package: `com.example.intenteventadapterdemo`): The receiving activity that initializes a `ListView` and populates it with subject strings using an `ArrayAdapter<String>`.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="30dp">

    <TextView
        android:id="@+id/txtTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Intent, Events and Listeners"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp" />

    <Button
        android:id="@+id/btnOpen"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Second Activity" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.intenteventadapterdemo;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnOpen;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnOpen = findViewById(R.id.btnOpen);

        btnOpen.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            startActivity(intent);
        });
    }
}
```

3. res/layout/activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Student Subjects"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginBottom="20dp" />

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

4. SecondActivity.java

```
package com.example.intenteventadapterdemo;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    ListView listView;

    String[] subjects = {
        "DBMS",
        "Operating System",
        "Android Programming",
        "Computer Networks",
        "Java Programming",
        "Python"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        listView = findViewById(R.id.listView);

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
```

```
            subjects
        );

        listView.setAdapter(adapter);
    }
}
```

Practical 09: Services, Notifications, and Broadcast Receivers

Aim: Programs on Services, notification and broadcast receivers.

Activity & Component Roles:

- `MainActivity.java` (Package: `com.example.servicenotificationbroadcast`): Handles `POST_NOTIFICATIONS` runtime permission on Android 13+ (API 33), triggers foreground background services, and dispatches explicit broadcast intents with action `"MY_CUSTOM_BROADCAST"`.
- `MyService` (Inner static class extending `Service`): Creates a `NotificationChannel` (required for Android 8.0+), runs as a foreground service with a persistent notification, and manages service lifecycle.
- `MyReceiver` (Inner static class extending `BroadcastReceiver`): Catches the broadcast event and informs the user via a Toast.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="30dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Services, Notification and Broadcast"
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginBottom="40dp" />

    <Button
        android:id="@+id/btnStartService"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="START SERVICE" />

    <Button
        android:id="@+id/btnStopService"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="STOP SERVICE"
        android:layout_marginTop="15dp" />
```

```
    <Button
        android:id="@+id/btnBroadcast"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SEND BROADCAST"
        android:layout_marginTop="15dp" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.servicenotificationbroadcast;

import android.Manifest;
import android.app.Notification;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.Service;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.os.IBinder;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private Button btnStartService;
    private Button btnStopService;
    private Button btnBroadcast;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            if (checkSelfPermission(Manifest.permission.POST_NOTIFICATIONS)
                    != PackageManager.PERMISSION_GRANTED) {
                requestPermissions(new String[]{Manifest.permission.POST_NOTIFICATIONS},
100);
            }
        }

        btnStartService = findViewById(R.id.btnStartService);
        btnStopService = findViewById(R.id.btnStopService);
        btnBroadcast = findViewById(R.id.btnBroadcast);

        btnStartService.setOnClickListener(v -> {
            Intent serviceIntent = new Intent(MainActivity.this, MyService.class);
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
                startForegroundService(serviceIntent);
            } else {
                startService(serviceIntent);
```

```
            }
            Toast.makeText(MainActivity.this, "Service Started",
Toast.LENGTH_SHORT).show();
        });

        btnStopService.setOnClickListener(v -> {
            Intent serviceIntent = new Intent(MainActivity.this, MyService.class);
            stopService(serviceIntent);
            Toast.makeText(MainActivity.this, "Service Stopped",
Toast.LENGTH_SHORT).show();
        });

        btnBroadcast.setOnClickListener(v -> {
            Intent broadcastIntent = new Intent(MainActivity.this, MyReceiver.class);
            broadcastIntent.setAction("MY_CUSTOM_BROADCAST");
            sendBroadcast(broadcastIntent);
        });
    }

    public static class MyService extends Service {

        private static final String CHANNEL_ID = "MY_SERVICE_CHANNEL";
        private static final int NOTIFICATION_ID = 1;

        @Override
        public void onCreate() {
            super.onCreate();
            createNotificationChannel();

            Notification notification;
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
                notification = new Notification.Builder(this, CHANNEL_ID)
                        .setContentTitle("Service Running")
                        .setContentText("My Android Service is running")
                        .setSmallIcon(android.R.drawable.ic_dialog_info)
                        .setOngoing(true)
                        .build();
            } else {
                notification = new Notification.Builder(this)
                        .setContentTitle("Service Running")
                        .setContentText("My Android Service is running")
                        .setSmallIcon(android.R.drawable.ic_dialog_info)
                        .setOngoing(true)
                        .build();
            }

            startForeground(NOTIFICATION_ID, notification);
        }

        @Override
        public int onStartCommand(Intent intent, int flags, int startId) {
            return START_STICKY;
        }

        @Override
        public void onDestroy() {
            super.onDestroy();
            NotificationManager manager = (NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
```

```
            if (manager != null) {
                manager.cancel(NOTIFICATION_ID);
            }
        }

        @Override
        public IBinder onBind(Intent intent) {
            return null;
        }

        private void createNotificationChannel() {
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
                NotificationChannel channel = new NotificationChannel(
                        CHANNEL_ID,
                        "My Service Notification",
                        NotificationManager.IMPORTANCE_DEFAULT
                );
                channel.setDescription("Notification for My Service");

                NotificationManager manager =
getSystemService(NotificationManager.class);
                if (manager != null) {
                    manager.createNotificationChannel(channel);
                }
            }
        }
    }

    public static class MyReceiver extends BroadcastReceiver {

        @Override
        public void onReceive(Context context, Intent intent) {
            if ("MY_CUSTOM_BROADCAST".equals(intent.getAction())) {
                Toast.makeText(context, "Broadcast Received Successfully!",
Toast.LENGTH_LONG).show();
            }
        }
    }
}
```

Practical 10: Database Programming with SQLite

Aim: Database Programming with SQLite.

Activity Role: `MainActivity.java` (Package: `com.example.sqlitedemo`) interacts directly with SQLite: opens/creates local database `StudentDB`, executes DDL `CREATE TABLE IF NOT EXISTS`, binds parameterized insert queries on INSERT click, and queries all student records with `rawQuery()` to display them formatted in an `AlertDialog` on VIEW click.

1. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
```

```
    android:padding="20dp">

    <EditText
        android:id="@+id/e1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter ID" />

    <EditText
        android:id="@+id/e2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name" />

    <EditText
        android:id="@+id/e3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Course" />

    <Button
        android:id="@+id/b1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="INSERT" />

    <Button
        android:id="@+id/b2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="VIEW" />

</LinearLayout>
```

2. MainActivity.java

```
package com.example.sqlitedemo;

import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText e1, e2, e3;
    SQLiteDatabase db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        e1 = findViewById(R.id.e1);
```

```
        e2 = findViewById(R.id.e2);
        e3 = findViewById(R.id.e3);

        Button b1 = findViewById(R.id.b1);
        Button b2 = findViewById(R.id.b2);

        db = openOrCreateDatabase("StudentDB", MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS student(id INTEGER, name TEXT, course
TEXT)");

        b1.setOnClickListener(v -> {
            db.execSQL(
                "INSERT INTO student VALUES(?,?,?)",
                new Object[]{
                    e1.getText().toString(),
                    e2.getText().toString(),
                    e3.getText().toString()
                }
            );
            Toast.makeText(this, "Data Inserted", Toast.LENGTH_SHORT).show();
        });

        b2.setOnClickListener(v -> {
            Cursor c = db.rawQuery("SELECT * FROM student", null);
            StringBuilder s = new StringBuilder();

            while (c.moveToNext()) {
                s.append("ID: ").append(c.getString(0))
                 .append("\nName: ").append(c.getString(1))
                 .append("\nCourse: ").append(c.getString(2))
                 .append("\n\n");
            }
            c.close();

            new AlertDialog.Builder(this)
                .setTitle("Student Records")
                .setMessage(s.toString())
                .setPositiveButton("OK", null)
                .show();
        });
    }
}
```